Bruce Bai

Fremont, California | brucebai002@gmail.com | (510) 409-5811 | github.com/bruceb002 | linkedin.com/in/bruce-bai/

WORK EXPERIENCE

Software Engineering Intern *June 2026 – August 2026*

Medtronic

- Developed a C++ GUI Proof of Concept to assess the feasibility of migrating embedded Linux from Chromium to WebKitGTK for engineer stakeholders, resolving UI errors while benchmarking CPU/memory parity with hardware acceleration enabled.
- Architected a simulated debug port framework Proof of Concept by integrating backend C++ Crow APIs and engineering complementary file scripts to mock automated Playwright frontend actions.
- Leveraged GitHub Copilot AI tool to trace complex system constraints across the stack, mapping dependencies between Yocto builds, Crow/Protobuf components, and React frontends to accelerate feature onboarding.

Software Engineering Intern *June 2025 – August 2025*

Olympiah Startup

- Diagnosed a homepage bottleneck across React, Node.js, and MongoDB by tracing request flows and removing unused geolocation indexing, reducing load times from up to 15 minutes to sub-second performance.
- Refactored REST APIs from Next.js routes to Express for MongoDB, validating functionality through Postman testing.

Software Engineering Intern *June 2023 – September 2023*

Best Egg Financial Services

- Integrated chatbot UI widget using Angular with TypeScript, configuring metadata handling for responsive user support.
- Prevented frontend regressions by authoring Karma unit tests and using spies with mock data fixtures in a sandbox.
- Merged verified features into production via Git pull requests, clearing automated CI/CD test gates for user support.

Financial Technology Intern *June 2022 - September 2022*

Marlette Funding LLC

- Cut forecasting runtimes from 12+ hours to 25 minutes by migrating a legacy SAS workflow to optimized SQL pipelines.
- Tuned relational database query performance using explain plans, strategic table indexing, and isolated SQL stored procedures.
- Enhanced data integrity by auditing table schemas for discrepancies and writing SQL logic to execute data cleansing rules.

SKILLS

Languages:	Python, C++, C, Java, JavaScript, TypeScript, SQL
Web Technologies:	React, Angular, Vue, HTML, CSS, Tailwind CSS, Express.js, Node.js, Flask, Prisma
Databases:	MongoDB, SQL Server, Snowflake, Cloud Firestore (NoSQL), Firebase Cloud Storage
Data & Machine Learning:	NumPy, Pandas, Scikit-learn
Developer Tools:	Git, Bitbucket, GitHub Actions, Docker, Postman, Jira, Linux/Unix, Vite
Development Practices:	Agile, Scrum, REST APIs, Unit Testing, SDLC

EDUCATION

University of Southern California

M.S. in Computer Science *Expected December 2026*

- Relevant Coursework: Artificial Intelligence, Machine Learning, Database Systems, Analysis of Algorithms, Computer Security

University of California Santa Cruz

B.S. in Computer Science, Cum Laude, GPA: 3.83 *September 2020 – June 2024*

- Relevant Coursework: Distributed Systems, Computer Systems Design, Computer Networking

PROJECTS

Floorplan Processing Optimization for Tactile Maps (Graduate Project) *September 2024 – April 2025*

- Parsed deeply nested JSON spatial datasets to isolate layout attributes (building types, wall structures, coordinates), navigating complex multi-index schemas to extract core geometric inputs.
- Isolated rendering anomalies by stress-testing TSP variants, evaluating Dynamic Programming, Simulated Annealing, and Ant Colony paradigms to pinpoint a root-cause failure in legacy wall-merging coordinate tracking.

Blackboard Mobile App (Capstone) *April 2024 – June 2024*

- Developed core React Native screens and custom UI components via Expo Go, including dynamic class registration flows and user metadata logic.
- Integrated Cloud Firestore and Firebase Cloud Storage to model NoSQL schemas for academic schedules and securely map user profile media uploads.
- Led software team through Agile sprints, using burn charts and Jira to track dependencies and deliver milestones.

JBS Rideshare Web Application *April 2023*

FIRST PLACE: 2023 ACM Hackathon

- Engineered Vue.js transit landing view, integrating Firebase APIs to map asynchronous driver data into structured UI blocks.